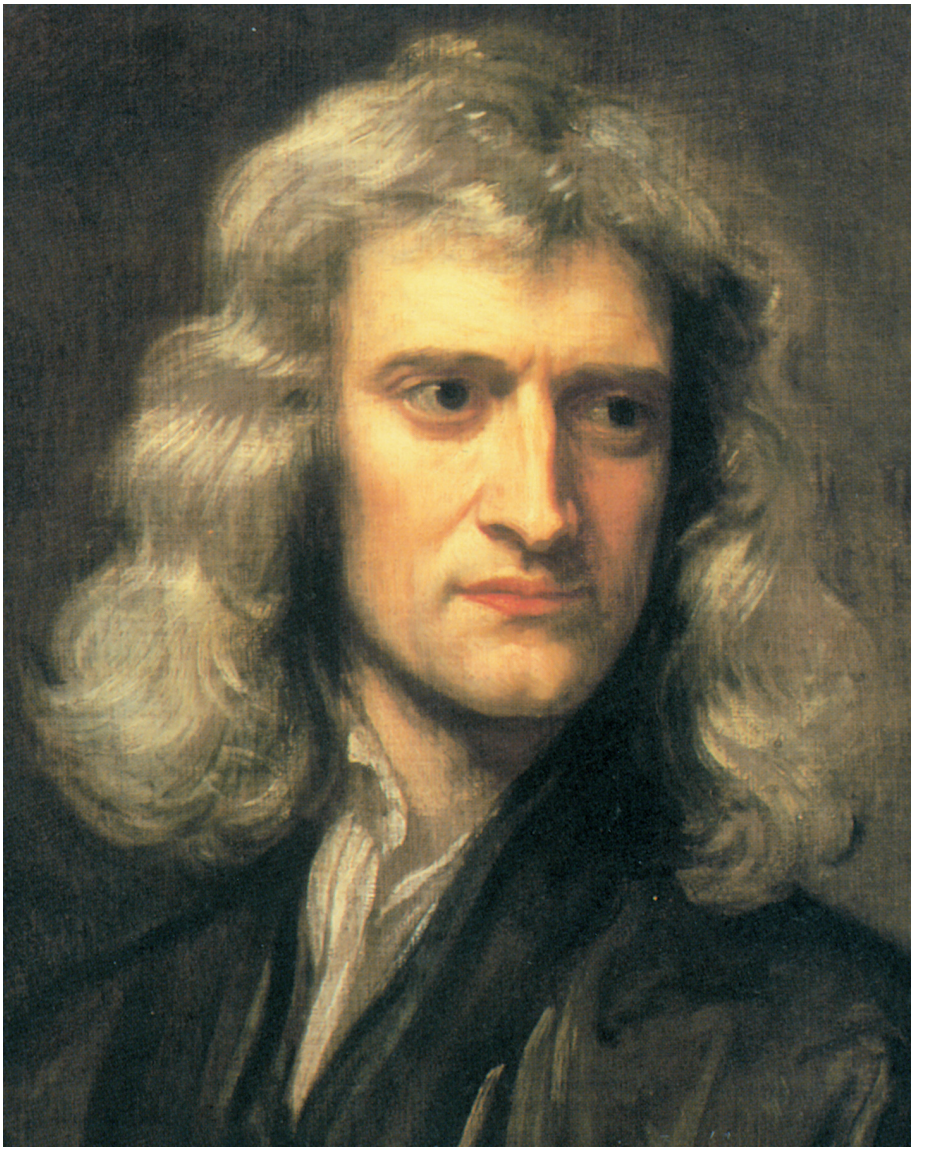
Part II

Newton



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Isaac Newton (1642–1727) had a rather tough childhood. His father died during his mother's pregnancy and his mother remarried when he was three years old—and left little Isaac in the care of his grandmother.

In 1661, Newton entered Trinity College in Cambridge, and graduated with a BA in 1664, after an unimpressive student career. But then the university shut down for two years because of the Great Plague, and Newton, back in his native Woolsthorpe, laid the ground for much of his future work in the *anni mirabiles* 1664–1666. He invented calculus (his method of *fluxions*) and the law of gravitation, and showed by experiment the prismatic composition of white light. All this before he turned 25. (*Inventing calculus* means that he developed a widely applicable theory; its roots go back, of course, to the work of many people, Archimedes and Fermat among them.)

Back at Cambridge, Newton became Lucasian Professor of Mathematics, at the age of 26. His former teacher, Isaac Barrow, resigned from that position to make way for the greater scientist (and to prepare his own move into a better position as chaplain to King Charles II). At that time, Newton was the prototype of the "forgetful professor", rather negligent about trifles such as his appearance. His nephew Humphrey Newton wrote: *He very rarely went to Dine in ye Hall unless upon some Publick Dayes, & then, if He has not been minded, would go very carelesly, wth Shooes down at Heels, Stockins unty'd, surplice on, & his Head scarcely comb'd.*

Newton did not publish his early results; this was later to work against him in disputes over priority. This was partly due to the publishers who were reluctant to invest in a money pit like a mathematical monograph. A 1672 paper on optics was so heavily criticized by overbearing referees (against whose beliefs his new and correct theory ran) that he withdrew it in the end.

Finally, he published in 1687 his masterpiece *Philosophiae Naturalis Principia Mathematica*, containing his discoveries in mechanics and astronomy.

In the summer of 1669, Newton had finished his *De Analysi per Æquationes Numero Terminorum Infinitas*. It circulated among English mathematicians and also abroad (Scotland and France), but appeared in print only in 1711. Among other things, he describes what is now called *Newton's method* (or Newton iteration) for approximating real roots of polynomial equations. He takes $\varphi = y^3 - 2y - 5 \in \mathbb{Q}[y]$ as an example, and proceeds as follows:

> *Let the Equation* $y^3 - 2y - 5 = 0$ *be proposed to be resolved: and let* 2 *be a number which differs from the Root sought, by less than a tenth Part of itself. Then I put* $2 + p = y$, *and I substitute this Value in Place of it in the Equation, and thence a new Equation arises,* viz. $p^3 + 6p^2 + 10p - 1 = 0$, *whose Root* p *is to be sought for, that it may*

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be added to the Quotient: viz. *thus (neglecting* $p^3 + 6p^2$ *upon the Account of their smallness)* $10p - 1 = 0$, *or* $p = 0.1$ *is near the Truth; therefore I write* 0.1 *in the Quotient, and then suppose* $0.1 + q = p$, *and this it's value I substitute, as formerly, whence results* $q^3 + 6.3q^2 + 11.23q + 0.061 = 0$.

And since $11.23q + 0.061 = 0$ *comes near to the Truth, or since* q *is almost equal to* -0.0054 *(*viz. *by dividing until as many Figures arise as there are places betwixt the first Figures of this and the principal Quotient) I write* -0.0054 *in the lower Part of the Quotient, since it is negative.*

His choice of the starting point 2 is justified by the fact that $\varphi(2) = -1 < 0 < 16 = \varphi(3)$, so that there is a root of φ between 2 and 3. This example later became a standard test for root finding methods. Joseph Raphson (1690) discussed this approach, acknowledging Newton as the originator, and it is sometimes called the Newton–Raphson method. Newton himself calls it "an improved version of the procedure, expounded by Viète and simplified by Oughtred". We will use Newton iteration in Chapters 9 and 15.



After many years of studying religious subjects (in particular, the Biblical chronology), Newton turned to public office, serving as Member of (powerless) Parliament until its dissolution in 1690. In 1699, he was awarded the moderately prestigious office of *Warden of the Mint.* Bell (1937) writes scathingly about this elevation: "The crowning imbecility of the Anglo-Saxon breed is its dumb belief in public office or an administrative position as the supreme honour for a man of intellect."—is that restricted to one "breed"?

The "universal genius" Leibniz, in Hannover, invented calculus independently, probably in the mid-1670s, and published this in 1684, before Newton published his ideas, almost two decades old by then. At first, the two men seem to have had mutual respect for each other's achievements. But fueled by the nationalism of the day (so, what else is new?), this degenerated into one of the bitterest controversies about priority in the history of science, an embarrassment for all persons involved.

Sir Isaac Newton, knighted by Queen Anne in 1705, was President of the Royal Society until his death at the age of 85.

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Classical in this context came to mean something like make-believe.
Richard Phillips Feynman (1984)

We shall not build a new world
until we have got rid of the mentalities of the old.
John le Carré (1989)

Rule 8: The development of fast algorithms is slow!
Arnold Schönhage (1994)

Jede mathematische Aufgabe könnte durch direktes Zählen gelöst werden. Es gibt aber Zähloperationen, die gegenwärtig in wenigen Minuten vollführt werden, welche aber ohne Methode vorzunehmen die Lebensdauer eines Menschen bei weitem nicht reichen würde.[1]
Ernst Mach (1896)

[1] Any mathematical task could, in principle, be solved by direct counting. However, there are counting problems that can presently be solved in a few minutes, but for which without mathematical method a lifetime would not be sufficient.

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